

# KharchaSplit Database

Data Model & Schema Reference

PostgreSQL · Backend package version: 1.3.0

Tables: 17 · ER diagram, columns, constraints, indexes

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# KharchaSplit Schema Reference

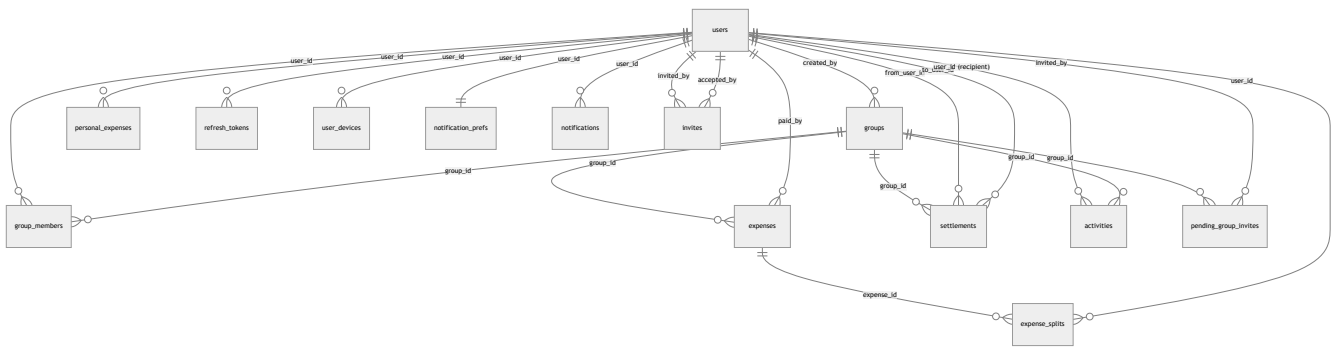
## KharchaSplit Backend — Data Model & Database Schema

PostgreSQL schema that backs the KharchaSplit Express API. Generated from the authoritative live source: `backend/src/config/initDatabase.js` (which runs at every server start) plus the historical migrations under `backend/migrations/`.

### 1. Conventions

| Concern      | Convention                                                                                                                                                             |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Engine       | PostgreSQL (uses <code>gen_random_uuid()</code> — <code>pgcrypto</code> / built-in 13+).                                                                               |
| Primary keys | <code>UUID</code> defaulted to <code>gen_random_uuid()</code> (older tables use <code>uuid_generate_v4()</code> ).                                                     |
| Time columns | <code>TIMESTAMP</code> (server time). Triggers maintain <code>updated_at</code> where present.                                                                         |
| Soft delete  | <code>deleted_at</code> <code>TIMESTAMP NULL</code> on most tables. <code>NULL</code> = live, non- <code>NULL</code> = gone.                                           |
| Booleans     | Backed by <code>BOOLEAN</code> (not <code>smallint</code> ). Defaults always set.                                                                                      |
| Currency     | <code>VARCHAR(10)</code> with default <code>'INR'</code> . ISO-4217 codes only (no symbols).                                                                           |
| Money        | <code>DECIMAL(12, 2)</code> (older tables: <code>DECIMAL(15, 2)</code> ). Two-decimal precision.                                                                       |
| JSON         | <code>JSONB</code> everywhere we use document storage (activities/notifications).                                                                                      |
| Cascade      | Child rows that depend on a parent (group_members, expenses, splits, ...) use <code>ON DELETE CASCADE</code> . Settlements + invites cascade off <code>groups</code> . |
| Naming       | <code>snake_case</code> columns. App converts to <code>camelCase</code> at the controller layer.                                                                       |
| Phone format | E.164 — regex <code>^\+?[1-9]\d{1,14}\$</code> . App normalises to <code>+91XXXXXXXXXX</code> for India.                                                               |

## 2. ER diagram



## 3. Core entities

### 3.1 users

Profile + auth identity. One row per phone number. Supports a "placeholder" mode where a row exists for someone invited by phone but who hasn't installed the app yet — when they sign up, the placeholder is converted in place (so they inherit any group memberships + expense history) by

```
User.convertPlaceholderToReal()
```

| Column                            | Type         | Default                        | Notes                                                                                           |
|-----------------------------------|--------------|--------------------------------|-------------------------------------------------------------------------------------------------|
| <code>id</code> (PK)              | UUID         | <code>gen_random_uuid()</code> |                                                                                                 |
| <code>phone_number</code>         | VARCHAR(20)  | —                              | <b>UNIQUE NOT NULL</b> . E.164. <code>users_phone_number_key</code> .                           |
| <code>name</code>                 | VARCHAR(100) | NULL                           | Empty on a freshly created placeholder.                                                         |
| <code>email</code>                | VARCHAR(255) | NULL                           | Optional.                                                                                       |
| <code>profile_image_base64</code> | TEXT         | NULL                           | Base64-encoded PNG/JPG, no MIME prefix.                                                         |
| <code>preferred_currency</code>   | VARCHAR(10)  | <code>'INR'</code>             | User's display currency. ISO-4217.                                                              |
| <code>is_placeholder</code>       | BOOLEAN      | <code>FALSE</code>             | <code>TRUE</code> for phone-only stub rows awaiting registration.                               |
| <code>fcm_token</code>            | VARCHAR(255) | NULL                           | <b>Legacy</b> single-device token; new code uses <code>user_devices</code> . Kept for backfill. |
| <code>fcm_token_updated_at</code> | TIMESTAMP    | NULL                           | Legacy.                                                                                         |
| <code>created_at</code>           | TIMESTAMP    | <code>NOW()</code>             |                                                                                                 |
| <code>updated_at</code>           | TIMESTAMP    | <code>NOW()</code>             | Trigger-maintained (legacy schema only).                                                        |
| <code>deleted_at</code>           | TIMESTAMP    | NULL                           | Soft delete.                                                                                    |

## Indexes

| Name                                            | Definition                                                                                                        |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <code>users_phone_number_key</code><br>(UNIQUE) | <code>phone_number</code>                                                                                         |
| <code>idx_users_fcm_token</code>                | <code>(fcm_token) WHERE fcm_token IS NOT NULL</code>                                                              |
| <code>idx_users_phone_normalized</code>         | Functional: <code>RIGHT(REGEXP_REPLACE(phone_number, '^[^0-9]', '', 'g'), 10)</code> — for last-10-digit lookups. |

## Constraints

- `CHECK (phone_number ~ '^\+?[1-9]\d{1,14}$')` (legacy migration adds this; new code relies on validator).

## 3.2 groups

Expense-sharing groups. `created_by` is preserved even if the user later leaves. `simplified_debts` is on by default — when true, the balances API merges chains of debt (A→B→C becomes A→C).

| Column                          | Type                 | Default                        | Notes                                             |
|---------------------------------|----------------------|--------------------------------|---------------------------------------------------|
| <code>id</code> (PK)            | UUID                 | <code>gen_random_uuid()</code> |                                                   |
| <code>name</code>               | VARCHAR(100)         | —                              | <b>NOT NULL</b> , 2–255 chars via validator.      |
| <code>description</code>        | TEXT                 | NULL                           |                                                   |
| <code>created_by</code>         | UUID<br>FK→users(id) | —                              |                                                   |
| <code>cover_image_base64</code> | TEXT                 | NULL                           | Optional group photo (base64).                    |
| <code>currency</code>           | VARCHAR(10)          | <code>'INR'</code>             | Display currency for this group.                  |
| <code>default_currency</code>   | VARCHAR(10)          | <code>'INR'</code>             | Currency used when creating a new expense.        |
| <code>simplified_debts</code>   | BOOLEAN              | <code>TRUE</code>              | If true, balances API simplifies debt chains.     |
| <code>is_archived</code>        | BOOLEAN              | <code>FALSE</code>             | Hidden from active view but data preserved.       |
| <code>archived_at</code>        | TIMESTAMP            | NULL                           | When archived (added by migration 005).           |
| <code>created_at</code>         | TIMESTAMP            | <code>NOW()</code>             |                                                   |
| <code>updated_at</code>         | TIMESTAMP            | <code>NOW()</code>             |                                                   |
| <code>deleted_at</code>         | TIMESTAMP            | NULL                           | Soft delete (blocked by API if any balance != 0). |

## Notes

- Both `currency` and `default_currency` exist for legacy reasons; controllers read whichever is populated, with `INR` fallback.

## 3.3 `group_members`

Membership rows. Carries denormalized name/phone/email so an expense created against a placeholder member still has display text even if that user's row is later deleted. `UNIQUE(group_id, user_id)` prevents duplicate memberships.

| Column                    | Type                                        | Default                        | Notes                                                          |
|---------------------------|---------------------------------------------|--------------------------------|----------------------------------------------------------------|
| <code>id</code> (PK)      | UUID                                        | <code>gen_random_uuid()</code> |                                                                |
| <code>group_id</code>     | UUID FK→groups(id) <b>ON DELETE CASCADE</b> | —                              |                                                                |
| <code>user_id</code>      | UUID FK→users(id)                           | —                              |                                                                |
| <code>role</code>         | VARCHAR(20)                                 | <code>'member'</code>          | <code>'creator'   'admin'   'member'</code> .                  |
| <code>name</code>         | VARCHAR(100)                                | NULL                           | Snapshoted from <code>users.name</code> at add time.           |
| <code>phone_number</code> | VARCHAR(20)                                 | NULL                           | Snapshoted.                                                    |
| <code>email</code>        | VARCHAR(255)                                | NULL                           | Snapshoted.                                                    |
| <code>added_by</code>     | UUID                                        | NULL                           | The inviter's user id.                                         |
| <code>joined_at</code>    | TIMESTAMP                                   | <code>NOW()</code>             |                                                                |
| <code>left_at</code>      | TIMESTAMP                                   | NULL                           | Set when the user leaves (alongside <code>deleted_at</code> ). |
| <code>deleted_at</code>   | TIMESTAMP                                   | NULL                           | Soft delete.                                                   |

## Indexes

| Name                                    | Definition                                                                                                |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------|
| <code>UNIQUE (group_id, user_id)</code> | also used for fast <code>group_id</code> prefix lookups.                                                  |
| <code>idx_group_members_user</code>     | <code>(user_id)</code> — for "my groups" lookups.                                                         |
| <code>idx_group_members_active</code>   | <code>(group_id, user_id, role, joined_at) WHERE deleted_at IS NULL</code> — covering index for hot path. |

## 3.4 expenses

A single bill someone paid that's being split. The participant breakdown lives in `expense_splits` .

| Column                      | Type                                        | Default                        | Notes                                                                                                                      |
|-----------------------------|---------------------------------------------|--------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| <code>id</code> (PK)        | UUID                                        | <code>gen_random_uuid()</code> |                                                                                                                            |
| <code>group_id</code>       | UUID FK→groups(id) <b>ON DELETE CASCADE</b> | —                              |                                                                                                                            |
| <code>description</code>    | VARCHAR(255)                                | —                              | <b>NOT NULL</b> . 1–500 chars enforced by validator.                                                                       |
| <code>amount</code>         | DECIMAL(12,2)                               | —                              | Non-negative. Validator enforces <code>amount &gt;= 0</code> .                                                             |
| <code>currency</code>       | VARCHAR(10)                                 | <code>'INR'</code>             |                                                                                                                            |
| <code>paid_by</code>        | UUID FK→users(id)                           | NULL                           | Current payer column. Read everywhere.                                                                                     |
| <code>paid_by_id</code>     | UUID FK→users(id)                           | NULL                           | <b>Legacy</b> . Was NOT NULL; relaxed by migration 012. Backfilled to <code>paid_by</code> .                               |
| <code>paid_by_name</code>   | VARCHAR(255)                                | NULL                           | <b>Legacy</b> denormalized name. Relaxed by migration 012.                                                                 |
| <code>category</code>       | VARCHAR(50)                                 | NULL                           | Free-form (e.g. <code>food</code> , <code>travel</code> ).                                                                 |
| <code>expense_date</code>   | DATE                                        | <code>CURRENT_DATE</code>      | Business date for the expense (not row create time).                                                                       |
| <code>split_type</code>     | VARCHAR(20)                                 | <code>'equal'</code>           | <code>'equal'</code>   <code>'unequal'</code>   <code>'exact'</code>   <code>'percentage'</code>   <code>'shares'</code> . |
| <code>receipt_base64</code> | TEXT                                        | NULL                           | Up to ~10 MB body limit per request.                                                                                       |
| <code>notes</code>          | TEXT                                        | NULL                           |                                                                                                                            |
| <code>is_deleted</code>     | BOOLEAN                                     | <code>FALSE</code>             | Legacy delete flag. New code uses <code>deleted_at</code> .                                                                |
| <code>created_at</code>     | TIMESTAMP                                   | <code>NOW()</code>             |                                                                                                                            |
| <code>updated_at</code>     | TIMESTAMP                                   | <code>NOW()</code>             |                                                                                                                            |
| <code>deleted_at</code>     | TIMESTAMP                                   | NULL                           | Soft delete.                                                                                                               |

## Indexes

| Name                                   | Definition                                                                                       |
|----------------------------------------|--------------------------------------------------------------------------------------------------|
| <code>idx_expenses_paid_by</code>      | <code>(paid_by)</code>                                                                           |
| <code>idx_expenses_group_active</code> | <code>(group_id, expense_date DESC, created_at DESC) WHERE deleted_at IS NULL</code> — hot path. |

## Authorization

- Only the original payer ( `paid_by` ) may edit or delete an expense.

### 3.5 `expense_splits`

One row per participant per expense. The sum of `amount` across an expense's splits should equal the expense `amount` . (The Flutter app guarantees this; the DB doesn't enforce it.)

| Column                  | Type                                                   | Default                        | Notes                                                      |
|-------------------------|--------------------------------------------------------|--------------------------------|------------------------------------------------------------|
| <code>id</code> (PK)    | UUID                                                   | <code>gen_random_uuid()</code> |                                                            |
| <code>expense_id</code> | UUID FK→expenses(id)<br><code>ON DELETE CASCADE</code> | —                              |                                                            |
| <code>user_id</code>    | UUID FK→users(id)                                      | —                              |                                                            |
| <code>amount</code>     | DECIMAL(12,2)                                          | —                              | <b>NOT NULL</b> . The actual money owed by this user.      |
| <code>percentage</code> | DECIMAL(5,2)                                           | NULL                           | Only populated when <code>split_type='percentage'</code> . |
| <code>shares</code>     | INTEGER                                                | NULL                           | Only populated when <code>split_type='shares'</code> .     |
| <code>is_settled</code> | BOOLEAN                                                | <code>FALSE</code>             | Set true when this split is covered by a settlement.       |
| <code>created_at</code> | TIMESTAMP                                              | <code>NOW()</code>             |                                                            |
| <code>deleted_at</code> | TIMESTAMP                                              | NULL                           |                                                            |

## Indexes

| Name                                  | Definition                                                              |
|---------------------------------------|-------------------------------------------------------------------------|
| <code>idx_expense_splits_batch</code> | <code>(expense_id, amount DESC)</code> — covers single + batch lookups. |
| <code>idx_expense_splits_user</code>  | <code>(user_id)</code>                                                  |

### 3.6 `settlements`

A pairwise payment that reduces outstanding debt in a group. Created by the payer, confirmed by the recipient (two-step). Affects balances exactly like an expense in reverse.



| Column                    | Type                                        | Default                        | Notes                                                                    |
|---------------------------|---------------------------------------------|--------------------------------|--------------------------------------------------------------------------|
| <code>id</code> (PK)      | UUID                                        | <code>gen_random_uuid()</code> |                                                                          |
| <code>group_id</code>     | UUID FK→groups(id) <b>ON DELETE CASCADE</b> | —                              |                                                                          |
| <code>from_user_id</code> | UUID FK→users(id)                           | —                              | The payer. Must equal <code>req.user.id</code> on create.                |
| <code>to_user_id</code>   | UUID FK→users(id)                           | —                              | The recipient. Only this user may confirm.                               |
| <code>amount</code>       | DECIMAL(12,2)                               | —                              | Non-negative.                                                            |
| <code>currency</code>     | VARCHAR(10)                                 | <code>'INR'</code>             |                                                                          |
| <code>status</code>       | VARCHAR(20)                                 | <code>'pending'</code>         | <code>'pending'   'confirmed'   'paid'   'failed'   'cancelled' .</code> |
| <code>notes</code>        | TEXT                                        | NULL                           |                                                                          |
| <code>confirmed_at</code> | TIMESTAMP                                   | NULL                           | Set when <code>to_user_id</code> confirms.                               |
| <code>settled_at</code>   | TIMESTAMP                                   | <code>NOW()</code>             | Initial create time (semantically separate from confirmed).              |
| <code>created_at</code>   | TIMESTAMP                                   | <code>NOW()</code>             |                                                                          |
| <code>deleted_at</code>   | TIMESTAMP                                   | NULL                           |                                                                          |

## Indexes

| Name                                      | Definition                                                                                                         |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| <code>idx_settlements_group_active</code> | <code>(group_id, created_at DESC) WHERE deleted_at IS NULL</code>                                                  |
| <code>idx_settlements_pending</code>      | <code>(group_id, from_user_id, to_user_id) WHERE status='pending' AND deleted_at IS NULL</code> — duplicate guard. |

## Constraints

- `CHECK (from_user_id != to_user_id)` — can't pay yourself.

## Balance math

The dashboard / group balance queries union expense contributions **and** settlement contributions, so a confirmed settlement immediately reduces the outstanding number for both users. Settlements with `status IN ('failed', 'cancelled')` are excluded.

## 3.7 `personal_expenses`

Private to one user; never visible to anyone else. No group, no splits.

| Column                      | Type              | Default                        | Notes                                           |
|-----------------------------|-------------------|--------------------------------|-------------------------------------------------|
| <code>id</code> (PK)        | UUID              | <code>gen_random_uuid()</code> |                                                 |
| <code>user_id</code>        | UUID FK→users(id) | —                              | Owner.                                          |
| <code>description</code>    | VARCHAR(255)      | —                              | <b>NOT NULL</b> , 1–500 chars.                  |
| <code>amount</code>         | DECIMAL(12,2)     | —                              | Non-negative.                                   |
| <code>currency</code>       | VARCHAR(10)       | <code>'INR'</code>             |                                                 |
| <code>category</code>       | VARCHAR(50)       | NULL                           |                                                 |
| <code>expense_date</code>   | TIMESTAMP         | <code>NOW()</code>             |                                                 |
| <code>receipt_base64</code> | TEXT              | NULL                           |                                                 |
| <code>notes</code>          | TEXT              | NULL                           |                                                 |
| <code>is_deleted</code>     | BOOLEAN           | <code>FALSE</code>             | Legacy. New code uses <code>deleted_at</code> . |
| <code>created_at</code>     | TIMESTAMP         | <code>NOW()</code>             |                                                 |
| <code>updated_at</code>     | TIMESTAMP         | <code>NOW()</code>             |                                                 |
| <code>deleted_at</code>     | TIMESTAMP         | NULL                           |                                                 |

## Indexes

| Name                                           | Definition                                                                                            |
|------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| <code>idx_personal_expenses_user_active</code> | <code>(user_id, expense_date DESC, created_at DESC) WHERE deleted_at IS NULL</code> — paginated list. |

## 3.8 activities

Activity-feed rows that appear in the user's bell-icon list. Written server-side by controllers via `ActivityService` (the `POST /activities` endpoint exists but is rarely used).

| Column                     | Type                                                 | Default                        | Notes                                                                                                                |
|----------------------------|------------------------------------------------------|--------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <code>id</code> (PK)       | UUID                                                 | <code>gen_random_uuid()</code> |                                                                                                                      |
| <code>user_id</code>       | UUID FK→users(id)                                    | —                              | Recipient.                                                                                                           |
| <code>group_id</code>      | UUID FK→groups(id)<br><code>ON DELETE CASCADE</code> | NULL                           | Optional — direct/user-scope activities have NULL.                                                                   |
| <code>activity_type</code> | VARCHAR(50)                                          | —                              | <b>NOT NULL</b> . e.g. <code>group_created</code> , <code>expense_added</code> , <code>settlement_confirmed</code> . |
| <code>entity_type</code>   | VARCHAR(50)                                          | NULL                           | <code>'group'</code>   <code>'member'</code>   <code>'expense'</code>   <code>'settlement'</code> .                  |
| <code>entity_id</code>     | UUID                                                 | NULL                           | FK target (not enforced).                                                                                            |
| <code>title</code>         | VARCHAR(255)                                         | NULL                           |                                                                                                                      |
| <code>description</code>   | TEXT                                                 | NULL                           |                                                                                                                      |
| <code>metadata</code>      | JSONB                                                | NULL                           | Free-form (amounts, names, etc.).                                                                                    |
| <code>is_read</code>       | BOOLEAN                                              | <code>FALSE</code>             |                                                                                                                      |
| <code>created_at</code>    | TIMESTAMP                                            | <code>NOW()</code>             |                                                                                                                      |
| <code>deleted_at</code>    | TIMESTAMP                                            | NULL                           |                                                                                                                      |

## Indexes

| Name                                   | Definition                                                                         |
|----------------------------------------|------------------------------------------------------------------------------------|
| <code>idx_activities_user_feed</code>  | <code>(user_id, created_at DESC) WHERE deleted_at IS NULL</code> — bell-icon feed. |
| <code>idx_activities_group_feed</code> | <code>(group_id, created_at DESC) WHERE deleted_at IS NULL</code>                  |
| <code>idx_activities_unread</code>     | <code>(user_id, is_read) WHERE deleted_at IS NULL AND is_read = FALSE</code>       |

## 4. Notification stack

### 4.1 `notification_prefs`

One row per user. Lazily created on first PUT — `GET` returns defaults without writing the row.

| Column                           | Type                                             | Default                        | Notes                                                  |
|----------------------------------|--------------------------------------------------|--------------------------------|--------------------------------------------------------|
| <code>user_id</code> (PK)        | UUID FK→users(id) <code>ON DELETE CASCADE</code> | —                              | One-to-one with <code>users</code> .                   |
| <code>push_enabled</code>        | BOOLEAN                                          | <code>TRUE</code>              | Master push switch.                                    |
| <code>email_enabled</code>       | BOOLEAN                                          | <code>FALSE</code>             | Email channel (not used yet).                          |
| <code>new_expense</code>         | BOOLEAN                                          | <code>TRUE</code>              | Per-type toggles below — each gates the matching push. |
| <code>group_invite</code>        | BOOLEAN                                          | <code>TRUE</code>              |                                                        |
| <code>payment_received</code>    | BOOLEAN                                          | <code>TRUE</code>              |                                                        |
| <code>settlement_reminder</code> | BOOLEAN                                          | <code>TRUE</code>              |                                                        |
| <code>comment_mention</code>     | BOOLEAN                                          | <code>TRUE</code>              |                                                        |
| <code>weekly_summary</code>      | BOOLEAN                                          | <code>FALSE</code>             |                                                        |
| <code>product_updates</code>     | BOOLEAN                                          | <code>FALSE</code>             |                                                        |
| <code>updated_at</code>          | TIMESTAMP                                        | <code>CURRENT_TIMESTAMP</code> |                                                        |

## 4.2 `user_devices`

Multi-device FCM token registry. Replaces the legacy `users.fcm_token` column (kept for backfill).

| Column                    | Type                                             | Default                        | Notes                                                                                               |
|---------------------------|--------------------------------------------------|--------------------------------|-----------------------------------------------------------------------------------------------------|
| <code>id</code> (PK)      | UUID                                             | <code>gen_random_uuid()</code> |                                                                                                     |
| <code>user_id</code>      | UUID FK→users(id) <code>ON DELETE CASCADE</code> | —                              |                                                                                                     |
| <code>fcm_token</code>    | TEXT                                             | —                              | <code>NOT NULL</code> . <code>UNIQUE</code> — makes the <code>ON CONFLICT</code> upsert idempotent. |
| <code>platform</code>     | VARCHAR(20)                                      | NULL                           | <code>'android'   'ios'   'web'</code> .                                                            |
| <code>device_name</code>  | VARCHAR(255)                                     | NULL                           |                                                                                                     |
| <code>os_version</code>   | VARCHAR(80)                                      | NULL                           |                                                                                                     |
| <code>app_version</code>  | VARCHAR(40)                                      | NULL                           |                                                                                                     |
| <code>last_seen_at</code> | TIMESTAMP                                        | <code>CURRENT_TIMESTAMP</code> | Bumped on every successful push delivery.                                                           |
| <code>created_at</code>   | TIMESTAMP                                        | <code>CURRENT_TIMESTAMP</code> |                                                                                                     |

## Indexes

| Name                                             | Definition                    |
|--------------------------------------------------|-------------------------------|
| <code>user_devices_fcm_token_key</code> (UNIQUE) | <code>( fcm_token )</code>    |
| <code>idx_user_devices_user_id</code>            | <code>( user_id )</code>      |
| <code>idx_user_devices_last_seen</code>          | <code>( last_seen_at )</code> |

### 4.3 notifications (inbox)

One row per delivered push. Persistent in-app inbox so the bell icon keeps history after the OS push is dismissed.

| Column                  | Type                                | Default                        | Notes                                                                                                    |
|-------------------------|-------------------------------------|--------------------------------|----------------------------------------------------------------------------------------------------------|
| <code>id</code> (PK)    | UUID                                | <code>gen_random_uuid()</code> |                                                                                                          |
| <code>user_id</code>    | UUID FK→users(id) ON DELETE CASCADE | —                              |                                                                                                          |
| <code>type</code>       | VARCHAR(50)                         | —                              | e.g. <code>EXPENSE_ADDED</code> ,<br><code>SETTLEMENT_REMINDER</code> ,<br><code>MEMBER_REMOVED</code> . |
| <code>title</code>      | TEXT                                | —                              |                                                                                                          |
| <code>body</code>       | TEXT                                | —                              |                                                                                                          |
| <code>data</code>       | JSONB                               | <code>'{}'</code>              | Deep-link payload (groupId, fromUserId, etc.).                                                           |
| <code>is_read</code>    | BOOLEAN                             | <code>FALSE</code>             |                                                                                                          |
| <code>read_at</code>    | TIMESTAMP                           | NULL                           |                                                                                                          |
| <code>created_at</code> | TIMESTAMP                           | <code>CURRENT_TIMESTAMP</code> |                                                                                                          |

## Indexes

| Name                                        | Definition                                                          |
|---------------------------------------------|---------------------------------------------------------------------|
| <code>idx_notifications_user_created</code> | <code>( user_id, created_at DESC )</code>                           |
| <code>idx_notifications_user_unread</code>  | <code>( user_id ) WHERE is_read = FALSE</code> — drives bell badge. |

## 5. Invitation tables

The app has **three** distinct invitation flows, each with its own table. (Yes, this is more than ideal — they grew organically.)

### 5.1 pending\_group\_invites

WhatsApp invites sent via WATI to non-registered phone numbers. A row here is paired with a `placeholder` user in `users`. Unique on `(group_id, phone_number)`.

| Column                       | Type                                        | Default                        | Notes                                                               |
|------------------------------|---------------------------------------------|--------------------------------|---------------------------------------------------------------------|
| <code>id</code> (PK)         | UUID                                        | <code>gen_random_uuid()</code> |                                                                     |
| <code>group_id</code>        | UUID FK→groups(id) <b>ON DELETE CASCADE</b> | —                              |                                                                     |
| <code>phone_number</code>    | VARCHAR(20)                                 | —                              | <b>NOT NULL</b> . E.164.                                            |
| <code>name</code>            | VARCHAR(100)                                | NULL                           |                                                                     |
| <code>email</code>           | VARCHAR(255)                                | NULL                           |                                                                     |
| <code>invited_by</code>      | UUID FK→users(id)                           | —                              |                                                                     |
| <code>wati_message_id</code> | VARCHAR(100)                                | NULL                           | WATI message handle for status tracking.                            |
| <code>wati_status</code>     | VARCHAR(20)                                 | <code>'pending'</code>         | <code>'pending'   'sent'   'delivered'   'read'   'failed'</code> . |
| <code>wati_error</code>      | TEXT                                        | NULL                           |                                                                     |
| <code>wati_sent_at</code>    | TIMESTAMP                                   | NULL                           |                                                                     |
| <code>invite_count</code>    | INTEGER                                     | <code>1</code>                 | Incremented on each resend.                                         |
| <code>last_invited_at</code> | TIMESTAMP                                   | <code>NOW()</code>             |                                                                     |
| <code>created_at</code>      | TIMESTAMP                                   | <code>NOW()</code>             |                                                                     |

### Indexes

| Name                                         | Definition                            |
|----------------------------------------------|---------------------------------------|
| <code>UNIQUE (group_id, phone_number)</code> | duplicate guard                       |
| <code>idx_pending_invites_group_phone</code> | <code>(group_id, phone_number)</code> |
| <code>idx_pending_invites_phone</code>       | <code>(phone_number)</code>           |

## 5.2 group\_invites

Group-scoped invite links (not currently exposed via the API — internal infrastructure for a future flow).

| Column          | Type                                 | Default           | Notes    |
|-----------------|--------------------------------------|-------------------|----------|
| id (PK)         | UUID                                 | gen_random_uuid() |          |
| group_id        | UUID FK→groups(id) ON DELETE CASCADE | —                 |          |
| invited_by      | UUID FK→users(id)                    | —                 |          |
| invited_phone   | VARCHAR(20)                          | NULL              |          |
| invited_user_id | UUID FK→users(id)                    | NULL              |          |
| status          | VARCHAR(20)                          | 'pending'         |          |
| invite_code     | VARCHAR(50)                          | NULL              | UNIQUE . |
| expires_at      | TIMESTAMP                            | NULL              |          |
| created_at      | TIMESTAMP                            | NOW()             |          |
| responded_at    | TIMESTAMP                            | NULL              |          |

## 5.3 invites (referral)

Share-sheet referral codes ( /api/v1/invites/\* ). Valid for 30 days.

| Column                    | Type                 | Default                        | Notes                                                            |
|---------------------------|----------------------|--------------------------------|------------------------------------------------------------------|
| <code>id</code> (PK)      | UUID                 | <code>gen_random_uuid()</code> |                                                                  |
| <code>invite_code</code>  | VARCHAR(20)          | —                              | <b>UNIQUE NOT NULL</b> . 8-char hex generated server-side.       |
| <code>invited_by</code>   | UUID<br>FK→users(id) | —                              |                                                                  |
| <code>phone_number</code> | VARCHAR(20)          | —                              | <b>NOT NULL</b> .                                                |
| <code>context</code>      | JSONB                | NULL                           | e.g. <code>{ "groupName": "..."} </code> — populates share text. |
| <code>status</code>       | VARCHAR(20)          | <code>'pending'</code>         | <code>'pending'   'accepted'   'expired'   'cancelled' </code> . |
| <code>accepted_by</code>  | UUID<br>FK→users(id) | NULL                           |                                                                  |
| <code>accepted_at</code>  | TIMESTAMP            | NULL                           |                                                                  |
| <code>expires_at</code>   | TIMESTAMP            | —                              | 30 days from create.                                             |
| <code>created_at</code>   | TIMESTAMP            | <code>NOW()</code>             |                                                                  |
| <code>deleted_at</code>   | TIMESTAMP            | NULL                           | Soft delete (used by Cancel).                                    |

## Indexes

| Name                                             | Definition                                                          |
|--------------------------------------------------|---------------------------------------------------------------------|
| <code>invites_invite_code_key</code><br>(UNIQUE) | <code>(invite_code)</code>                                          |
| <code>idx_invites_invited_by</code>              | <code>(invited_by, created_at DESC) WHERE deleted_at IS NULL</code> |
| <code>idx_invites_phone</code>                   | <code>(phone_number) WHERE deleted_at IS NULL</code>                |

## 6. Auth tables

### 6.1 `otps`

Short-lived OTP rows used by `/auth/register` . (The newer `/auth/send-otp`

- `/auth/verify-otp` flow uses **Twilio Verify** and does NOT write to this table — kept for the legacy register path.)



| Column                    | Type        | Default                        | Notes      |
|---------------------------|-------------|--------------------------------|------------|
| <code>id</code> (PK)      | UUID        | <code>gen_random_uuid()</code> |            |
| <code>phone_number</code> | VARCHAR(20) | —                              | NOT NULL . |
| <code>otp</code>          | VARCHAR(10) | —                              | NOT NULL . |
| <code>verified</code>     | BOOLEAN     | FALSE                          |            |
| <code>expires_at</code>   | TIMESTAMP   | —                              | NOT NULL . |
| <code>created_at</code>   | TIMESTAMP   | NOW()                          |            |

## Indexes

| Name                               | Definition                                                             |
|------------------------------------|------------------------------------------------------------------------|
| <code>idx_otps_phone_verify</code> | <code>(phone_number, verified, expires_at DESC)</code> — verification. |

## 6.2 `refresh_tokens`

One row per signed-in session. Used by the Active Sessions screen. Tokens expire after 30 days of inactivity ( `last_used_at` is bumped on every refresh). Migration 009 added the device-fingerprint columns.

| Column                    | Type                                             | Default                        | Notes                                        |
|---------------------------|--------------------------------------------------|--------------------------------|----------------------------------------------|
| <code>id</code> (PK)      | UUID                                             | <code>gen_random_uuid()</code> |                                              |
| <code>user_id</code>      | UUID FK→users(id) <code>ON DELETE CASCADE</code> | —                              |                                              |
| <code>token</code>        | TEXT                                             | —                              | <code>NOT NULL</code> . JWT.                 |
| <code>expires_at</code>   | TIMESTAMP                                        | —                              | <code>NOT NULL</code> . 30 days from create. |
| <code>device_name</code>  | VARCHAR(255)                                     | NULL                           | "Pixel 7", "iPhone 15".                      |
| <code>platform</code>     | VARCHAR(50)                                      | NULL                           | <code>'android'   'ios'   'web'</code> .     |
| <code>os_version</code>   | VARCHAR(80)                                      | NULL                           |                                              |
| <code>app_version</code>  | VARCHAR(40)                                      | NULL                           |                                              |
| <code>ip_address</code>   | VARCHAR(64)                                      | NULL                           |                                              |
| <code>user_agent</code>   | TEXT                                             | NULL                           |                                              |
| <code>last_used_at</code> | TIMESTAMP                                        | NULL                           | Bumped by <code>/auth/refresh</code> .       |
| <code>created_at</code>   | TIMESTAMP                                        | <code>NOW()</code>             |                                              |

Indexes

| Name                                      | Definition                       |
|-------------------------------------------|----------------------------------|
| <code>idx_refresh_tokens_user</code>      | <code>(user_id)</code>           |
| <code>idx_refresh_tokens_lookup</code>    | <code>(token, expires_at)</code> |
| <code>idx_refresh_tokens_last_used</code> | <code>(last_used_at)</code>      |

7. Sync (legacy)

7.1 `sync_metadata`

Per-user, per-table last-sync timestamp. The `POST /sync` endpoint exists but its handlers are stubs — the table is written but the operations are not yet replayed. Kept here for completeness.

| Column                                                           | Type                                       | Default                         | Notes           |
|------------------------------------------------------------------|--------------------------------------------|---------------------------------|-----------------|
| <code>id</code> (PK)                                             | UUID                                       | <code>uuid_generate_v4()</code> |                 |
| <code>user_id</code>                                             | UUID FK→users(id) <b>ON DELETE CASCADE</b> | —                               |                 |
| <code>table_name</code>                                          | VARCHAR(50)                                | —                               | <b>NOT NULL</b> |
| <code>last_synced_at</code>                                      | TIMESTAMP                                  | <code>NOW()</code>              |                 |
| <b>UNIQUE</b> ( <code>user_id</code> , <code>table_name</code> ) | —                                          | —                               |                 |

## 8. Enums / reserved values

Free-form `VARCHAR` columns with conventional values. No DB-level `ENUM` types are used.

| Column                                         | Allowed values                                                                                                                                                                                                                                                                                                                                                      |
|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>group_members.role</code>                | <code>'creator'</code> , <code>'admin'</code> , <code>'member'</code>                                                                                                                                                                                                                                                                                               |
| <code>expenses.split_type</code>               | <code>'equal'</code> , <code>'unequal'</code> , <code>'exact'</code> , <code>'percentage'</code> , <code>'shares'</code>                                                                                                                                                                                                                                            |
| <code>settlements.status</code>                | <code>'pending'</code> , <code>'confirmed'</code> , <code>'paid'</code> , <code>'failed'</code> , <code>'cancelled'</code>                                                                                                                                                                                                                                          |
| <code>invites.status</code>                    | <code>'pending'</code> , <code>'accepted'</code> , <code>'expired'</code> , <code>'cancelled'</code>                                                                                                                                                                                                                                                                |
| <code>pending_group_invites.wati_status</code> | <code>'pending'</code> , <code>'sent'</code> , <code>'delivered'</code> , <code>'read'</code> , <code>'failed'</code>                                                                                                                                                                                                                                               |
| <code>activities.activity_type</code>          | <code>'group_created'</code> , <code>'group_updated'</code> , <code>'group_deleted'</code> , <code>'group_archived'</code> , <code>'member_added'</code> , <code>'member_removed'</code> , <code>'expense_added'</code> , <code>'expense_updated'</code> , <code>'expense_deleted'</code> , <code>'settlement_created'</code> , <code>'settlement_confirmed'</code> |
| <code>notifications.type</code>                | <code>'EXPENSE_ADDED'</code> , <code>'SETTLEMENT_CREATED'</code> , <code>'SETTLEMENT_CONFIRMED'</code> , <code>'SETTLEMENT_REMINDER'</code> , <code>'GROUP_INVITE'</code> , <code>'GROUP_ARCHIVED'</code> , <code>'GROUP_COMPLETED'</code> , <code>'MEMBER_REMOVED'</code> , <code>'MEMBER_LEFT'</code>                                                             |
| <code>user_devices.platform</code>             | <code>'android'</code> , <code>'ios'</code> , <code>'web'</code>                                                                                                                                                                                                                                                                                                    |
| <code>refresh_tokens.platform</code>           | same as above                                                                                                                                                                                                                                                                                                                                                       |

## 9. Data lifecycle

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### 9.1 Placeholder users

1. Existing user invites a phone that has no account.
2. `User.createPlaceholder()` inserts a row with `is_placeholder = TRUE` and an empty `name`.
3. A `pending_group_invites` row is inserted; WATI WhatsApp invite goes out.
4. The placeholder is added as a real `group_members` row, so expenses can reference them.
5. When the real person registers → `User.convertPlaceholderToReal()` flips `is_placeholder = FALSE`, fills in `name` / `email`, and **deletes** the `pending_group_invites` row(s). The user keeps every group + expense they were added to.

### 9.2 Soft deletes

`deleted_at` is the modern flag. Older tables also have `is_deleted BOOLEAN` (still written by legacy code paths). All read queries filter `WHERE deleted_at IS NULL` — there's no restore API.

### 9.3 Group deletion guard

`DELETE /groups/:id` is **rejected** (HTTP 409) if any pair of members in the group has a non-zero balance. The check unions expense contributions + settlement contributions.

### 9.4 Member self-leave guard

Self-removal via `DELETE /groups/:id/members/:userId` is rejected (HTTP 409) if the caller has any unsettled pairwise debt with another group member. Admins removing someone else are not subject to this check.

### 9.5 Reminder rate limit (data-driven)

`POST /groups/:id/remind/:userId` queries `notifications` for any row in the last 6 hours with matching `(user_id=target, type='SETTLEMENT_REMINDER', data->>'fromUserId'=caller, data->>'groupId'=group)`. If found → HTTP 429.

### 9.6 Cache layer

Reads are cached in-process ( `backend/src/services/cacheService.js` ). Notable keys:

- `auth:user:<userId>` — every authenticated request consults this (5 min TTL).
- `group:<id>:balances`, `group:<id>:settlements`, `group:<id>:expenses` — invalidated by `Expense.invalidateGroupExpenses()` + settlement-controller `cache.invalidate(...)` on writes.

The cache is **not** distributed — restart the server and it warms up clean. Fine for a single-instance deploy; swap to Redis if scaling out.

---

## 10. Migration history

| #   | File                                                 | What it did                                                                                                                                                                        |
|-----|------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 001 | <code>001_initial_schema.sql</code>                  | Initial tables (users, groups, group_members, expenses, expense_participants*, settlements, personal_expenses, otps, refresh_tokens, sync_metadata, activities) + triggers + view. |
| 002 | <code>002_add_currency_to_groups.sql</code>          | <code>groups.currency VARCHAR(3) DEFAULT 'INR' .</code>                                                                                                                            |
| 003 | <code>003_add_preferred_currency_to_users.sql</code> | <code>users.preferred_currency VARCHAR(3) DEFAULT 'INR' .</code>                                                                                                                   |
| 004 | <code>004_add_invites_table.sql</code>               | Referral <code>invites</code> table.                                                                                                                                               |
| 004 | <code>004_fix_duplicate_phone_numbers.sql</code>     | One-off cleanup script for duplicate phone numbers.                                                                                                                                |
| 005 | <code>005_add_archived_at_to_groups.sql</code>       | <code>groups.archived_at TIMESTAMP .</code>                                                                                                                                        |
| 006 | <code>006_add_fcm_token_to_users.sql</code>          | <code>users.fcm_token</code> + <code>fcmtoken_updated_at</code> (legacy single-device token).                                                                                      |
| 007 | <code>007_add_pending_invites_table.sql</code>       | <code>pending_group_invites</code> table for WATI flow.                                                                                                                            |
| 008 | <code>008_add_placeholder_user_support.sql</code>    | <code>users.is_placeholder</code> + supporting indexes.                                                                                                                            |
| 009 | <code>009_add_device_info_to_sessions.sql</code>     | Device-fingerprint columns on <code>refresh_tokens</code> .                                                                                                                        |
| 010 | <code>010_add_notifications.sql</code>               | <code>notification_prefs</code> + <code>user_devices</code> (multi-device FCM).                                                                                                    |
| 011 | <code>011_add_notifications_inbox.sql</code>         | <code>notifications</code> inbox table.                                                                                                                                            |
| 012 | <code>012_relax_expenses_paid_by_legacy.sql</code>   | Dropped <code>NOT NULL</code> on legacy <code>expenses.paid_by_id</code> / <code>paid_by_name</code> ; backfilled <code>paid_by</code> .                                           |

\* `expense_participants` is in migration 001 but **not actually used at runtime** — `initDatabase.js` creates `expense_splits` instead, which is what every controller queries. The unused legacy table should be cleaned up in a future migration.

## 11. Quick table index

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|                       |                                       |
|-----------------------|---------------------------------------|
| users                 | (auth + profile)                      |
| groups                | (expense-sharing rooms)               |
| group_members         | (membership + role)                   |
| expenses              | (bills paid in a group)               |
| expense_splits        | (per-participant share)               |
| settlements           | (pairwise debt payments)              |
| personal_expenses     | (private to one user)                 |
| activities            | (in-app activity feed)                |
| notification_prefs    | (per-user push toggles)               |
| user_devices          | (multi-device FCM tokens)             |
| notifications         | (in-app inbox of delivered pushes)    |
| pending_group_invites | (WATI WhatsApp invites)               |
| group_invites         | (group-scoped invite codes; reserved) |
| invites               | (referral / share-sheet invites)      |
| otps                  | (legacy register OTPs)                |
| refresh_tokens        | (active sessions)                     |
| sync_metadata         | (last-synced timestamps; stub flow)   |